

CS6284: ADVANCED TOPICS IN SOFTWARE SECURITY

New Syllabus Proposal

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Advanced Topics in Software Security

Subject Code

CS - Computer Science

Course Number

6284

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS2117 Foundation of Cybersecurity
AND CS3103 Operating Systems
AND CS3104 Applied Cryptographic Systems
AND CS4101 Software Security

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course aims to provide students with a solid understanding of a range of topics in the field of software security principles, practices, and techniques, emphasizing the identification of security threats to real-world software systems and the appropriate countermeasures. Students are exposed to basic programming constructs (such as variables, control structures, data structures, programming syntax) as well as the basic principles of object-oriented programming languages. Upon completing the course, students will learn about common vulnerabilities, threat models, and secure coding practices. The course covers both theoretical concepts and practical applications, preparing students to design and develop secure software systems. Meanwhile, the students can also learn to specify and evaluate appropriate security measures for computer systems and software applications.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Analyse common threats and vulnerabilities of software, network, and system applications.	20	x	x	
2	Study and present major state-of-the-art countermeasures to software, network, and system attacks. Explore the strengths and weaknesses of existing techniques.	35	x	x	x
3	Implement or evaluate countermeasures techniques to demonstrate the understanding of existing techniques and improve the effectiveness of detecting and localizing vulnerability in real-world applications.	45	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Lectures introduce vulnerabilities in diverse application systems, such as software, AI, and web applications. Lectures demonstrate the defense principles, techniques, and technologies used for these attacks. Lectures additionally discuss selected timely security issues in emerging computing technology.	1, 2, 3	2 hours/ week
2	Tutorials	Tutorials are conducted in the laboratory through discussion, demonstration, and hands-on sessions. Students will work with selected security issues and attacking tools, which provide students with hands-on experience in using and configuring the tools and analyzing how the security and attacking tools work. With these exercises, students will learn how the adversary uses the tool to attack software and web applications. Students should be able to identify and analyze potential threats to computer systems in organizations and formulate solutions for how organizations may defend themselves.	1, 2, 3	1 hour/ week

3	Projects	Students will be asked to conduct a substantial case study or in-depth survey on selected security topics, such as thoroughly analyzing the security properties of crypto techniques in state-of-the-art system kernels, network protocols, advanced access control, passwords, and related usages, memory safety issues and defenses, web tracking, command injection attacks and defenses, cloud security, etc.	1, 2, 3	2 hours/ week for 4 weeks
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Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignment 1	1, 2	10	-	Yes
2	Assignment 2	1, 2, 3	10	-	Yes
3	Project	1, 2, 3	20	-	Yes

Continuous Assessment (%)

40

Examination (%)

60

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark of the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Assignments including problem set and hands-on exercises

Criterion

Ability to answer fundamental software security attacks and defences.

Excellent

(A+, A, A-): High

Good

(B+, B, B-): Significant

Fair

(C+, C, C-): Moderate

Marginal

(D): Basic

Failure

(F): Not even reaching marginal level

Assessment Task

Project

Criterion

Ability to conduct a substantial case study in vulnerability exploitation and in-depth survey on security topics.

Excellent

(A+, A, A-): High

Good

(B+, B, B-): Significant

Fair

(C+, C, C-): Moderate

Marginal

(D): Basic

Failure

(F): Not even reaching marginal level

Assessment Task

Examination

Criterion

The exam will include questions to assess the student's ability to explain the principles, techniques and technologies used for software security and its applications in practice.

Excellent

(A+, A, A-): High

Good

(B+, B, B-): Significant

Fair

(C+, C, C-): Moderate

Marginal

(D): Basic

Failure

(F): Not even reaching marginal level

Part III Other Information

Keyword Syllabus

The syllabus evolves over time as security topics change. The following is a list of potential course topics:

- 1) Security principle and threat model.
- 2) Software security and its countermeasures.
- 3) AI-related security and its countermeasures.
- 4) Mobile security and its countermeasures.
- 5) Secure coding practices.
- 6) Other emerging topics in computer security: Cloud security, information governance, information privacy, security evaluation, legal and ethical issues, computer crime and computer forensics, new access control paradigms, and database security.

Reading List

Compulsory Readings

Title	
1	Nil

Additional Readings

Title	
1	Gary McGraw. Software Security - Building Security
2	M. Goodrich and R. Tamassia. Introduction to Computer Security. Pearson. (2014)
3	W. Stallings and L. Brown. Computer Security: Principles and Practice. (2015)
4	Shah S. Web 2.0 security: Defending Ajax, RIA, and SOA. Thomson (2008)
5	Spitzner L. Honeypot: Tracking hackers. Addison-Wesley (2003)
6	Whittaker and Thompson. How to break software security. Addison Wesley (2004)
7	Bace R. G. Intrusion Detection. Macmillan Technical (2000)
8	Skoudis and Liston, Counter Hack Reloaded (2e). Prentice Hall (2006)
9	Jon Erickson. Hacking: The Art of Exploitation, 2nd Edition. No Starch Press, 2008
10	Andrews and Whittaker. How to break web software. Addison Wesley (2006)